

Employee Retention Prediction System

An End-to-End Supervised Machine Learning Pipeline for Attrition Risk Stratification and Prescriptive HR Interventions

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Domain: Machine Learning • People Analytics

Academic Evaluation: TAE-1 Project Based Learning

Core Algorithm: Cost-Sensitive Random Forest (Champion)

Compulsory Project Submission Links:

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GitHub Repository URL:

<https://github.com/JayeshGajbhiye/Employee-Retention-Prediction-System>

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Live Vercel Web Application:

<https://employee-retention-prediction-syste.vercel.app>

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Live API Endpoint:

<https://employee-retention-prediction-syste.vercel.app/api/predict>

1. Executive Summary & Problem Formulation

Human capital retention is among the most urgent challenges confronting modern organizations. Unplanned voluntary attrition triggers substantial direct financial loss, with industry studies from the Society for Human Resource Management (SHRM) demonstrating that replacing an employee costs between **100% and 200% of their annual salary**. Beyond direct monetary costs, turnover causes loss of institutional knowledge, disrupts project timelines, and increases burnout among retained staff.

Traditional Human Resources operations rely on reactive instruments, predominantly exit interviews performed after an employee has already tendered their resignation. This project develops and deploys an end-to-end predictive machine learning decision-support system that proactively calculates employee turnover probability ($P(\text{Attrition} = 1)$), detects underlying organizational friction points, and prescribes targeted retention actions before departures occur.

2. System Architecture & Workflow Pipeline

The system follows a modular production architecture comprising data cleaning, leak-free featurization, multi-algorithm training with cross-validation, hyperparameter optimization, and a dual deployment strategy (Streamlit and Vercel):

Pipeline Sequence: Raw IBM HR Dataset (1,470 samples, 35 features) → Zero-variance elimination → Stratified Train/Test Split (80/20) → Feature Engineering → Preprocessing Pipeline Fit (Train only) → Stratified 5-Fold Cross-Validation → Multi-Model Evaluation (Logistic Regression, Decision Tree, Random Forest, SVC, XGBoost) → GridSearchCV Hyperparameter Tuning → Holdout Evaluation → Interactive Vercel & Streamlit Cloud Deployment.

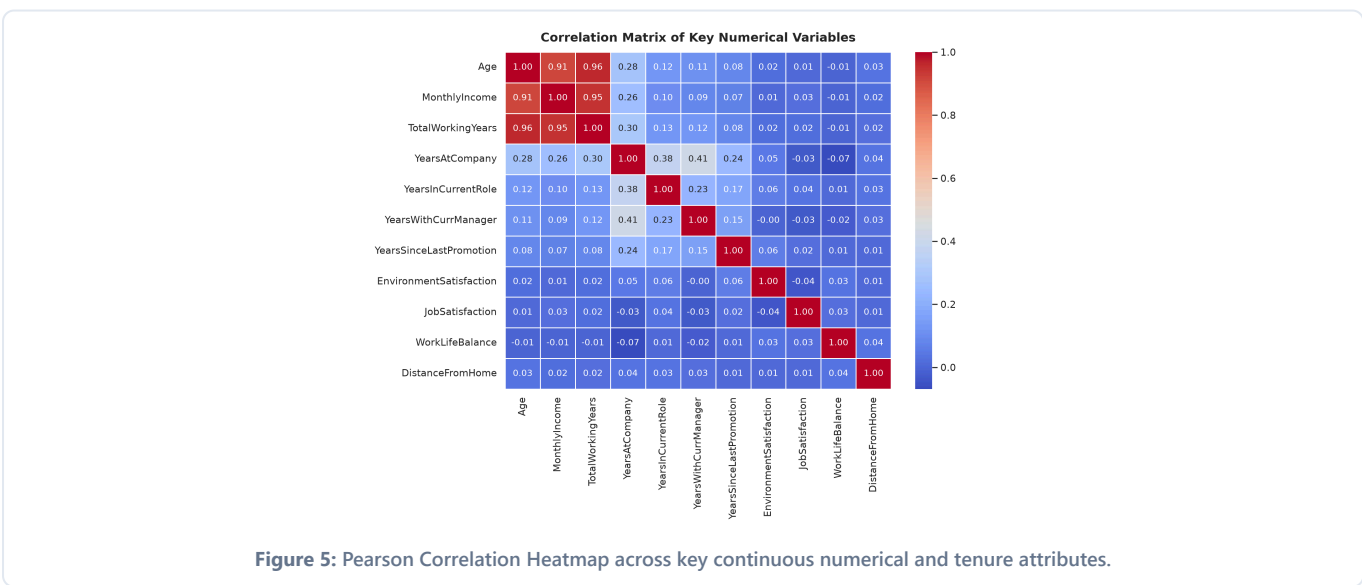
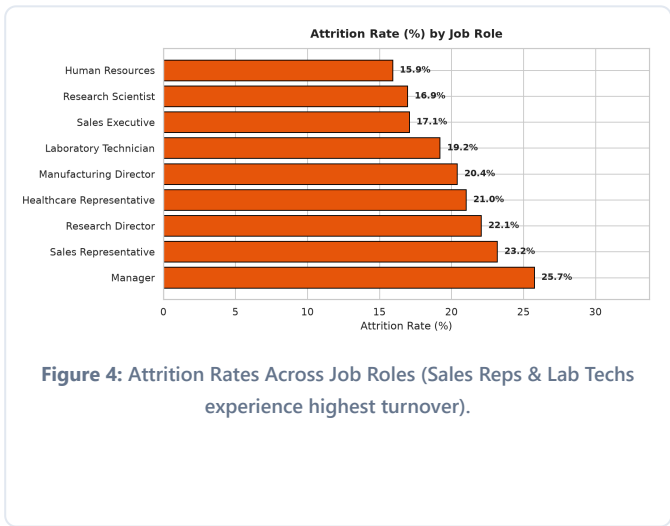
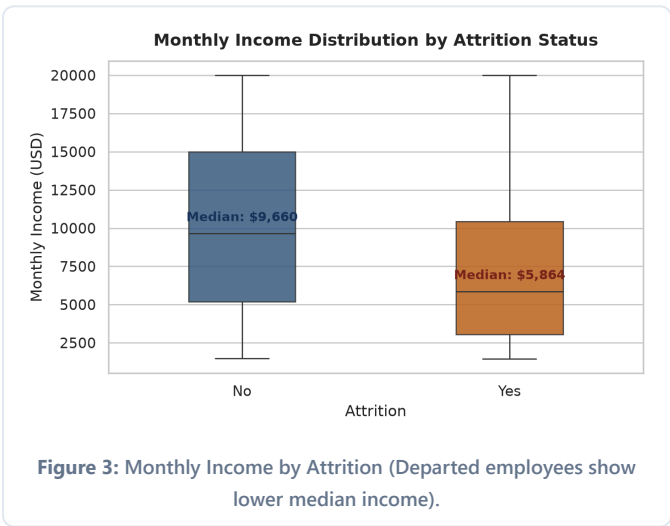
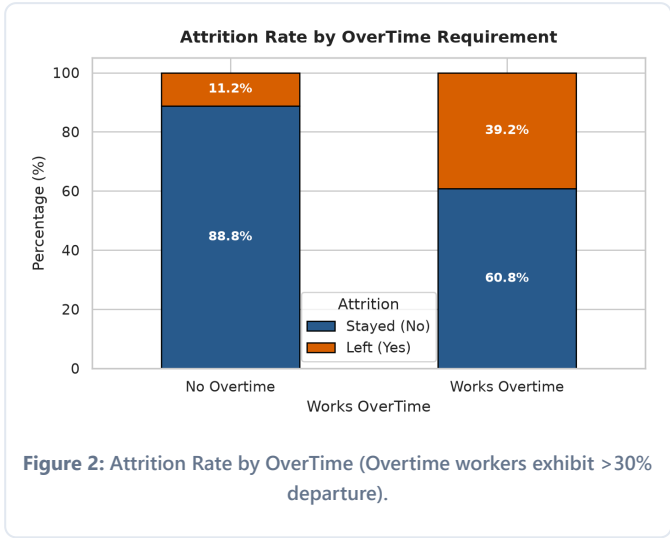
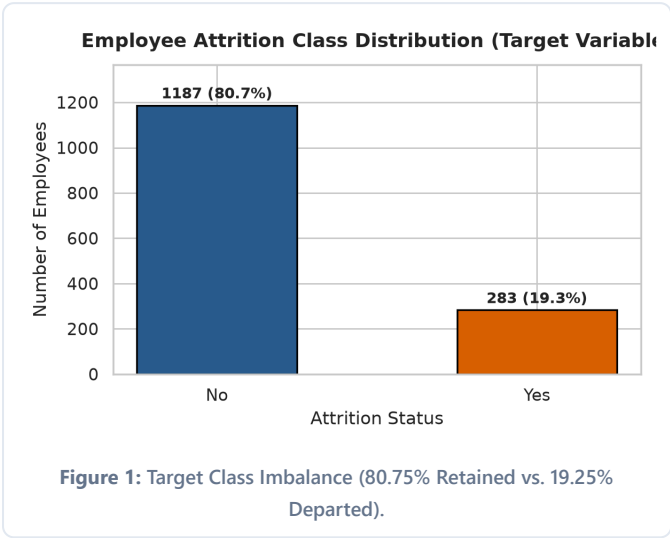
3. Dataset Description & Variable Dictionary

The project leverages the canonical **IBM HR Analytics Employee Attrition** benchmark dataset (\$N = 1,470\$ instances, 35 columns). Target variable Attrition is binary ("Yes" = 283, "No" = 1,187), reflecting an authentic industry turnover baseline of **19.25%**.

Variable Category	Key Features	Data Types	Operational Significance
Target Feature	Attrition	Binary (Yes / No)	Ground truth departure status (Positive class: 19.25%)
Demographics	Age, Gender, MaritalStatus, Education, EducationField	Numerical & Nominal	Career lifecycle maturity and biographical background
Organizational Role	Department, JobRole, JobLevel, BusinessTravel, OverTime	Nominal & Ordinal	Workload intensity, travel burdens, and managerial hierarchy
Compensation	MonthlyIncome, DailyRate, HourlyRate, MonthlyRate, PercentSalaryHike, StockOptionLevel	Continuous Numerical	Financial reward competitiveness and market alignment
Tenure & Career	TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsWithCurrManager, YearsSinceLastPromotion	Discrete Count	Career progression velocity, role stagnation, and tenure stability
Surveys & Morale	JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction, WorkLifeBalance, JobInvolvement	Ordinal Scale (1–4)	Subjective employee satisfaction and workplace sentiment
Constants / Metadata	EmployeeCount, StandardHours, Over18, EmployeeNumber	Zero-Variance	Dropped during preprocessing to prevent dimensionality bloat

4. Exploratory Data Analysis (EDA) & Key Findings

Comprehensive exploratory data analysis was conducted to uncover underlying statistical distributions and correlations driving voluntary employee turnover:



5. Preprocessing & Leak-Free Feature Engineering

Adhering strictly to enterprise Machine Learning best practices, all transformations adhered to **strict featurization ordering**:

- **Zero-Variance Dropping:** EmployeeCount, StandardHours, Over18, and EmployeeNumber were removed.
- **Stratified Train-Test Partitioning:** The dataset was split into 80% training ($N = 1,176$) and 20% independent holdout test ($N = 294$) sets prior to any scaling or encoding to guarantee complete isolation and prevent data leakage.
- **Engineered Domain Features:** Five domain-specific HR ratios were derived:

```
TenurePerJob = TotalWorkingYears / (NumCompaniesWorked + 1)
CompositeSatisfaction = (EnvironmentSatisfaction + JobSatisfaction + RelationshipSatisfaction +
                          WorkLifeBalance) / 4
YearsWithManagerRatio = YearsWithCurrManager / (YearsAtCompany + 1)
IncomePerWorkingYear = MonthlyIncome / (TotalWorkingYears + 1)
YearsWithoutPromotionRatio = YearsSinceLastPromotion / (YearsAtCompany + 1)
```

Numerical features were normalized via StandardScaler, and categorical variables encoded with OneHotEncoder(drop='first', handle_unknown='ignore'), expanding the feature space to 50 orthogonal model inputs.

6. Model Benchmark & Experimental Evaluation

Five diverse classification algorithms were evaluated alongside hyperparameter tuning. Cost-sensitive weighting was implemented across all models to mitigate class imbalance.

6.1 Stratified 5-Fold Cross-Validation Performance (Training Set)

Candidate Architecture	Mean CV ROC-AUC	Std CV ROC-AUC	Mean CV Recall	Mean CV F1	Mean CV Accuracy
Random Forest Classifier	0.8189	± 0.0286	0.5088	0.4925	79.93%
Logistic Regression (Balanced)	0.8050	± 0.0318	0.7036	0.5175	74.66%
XGBoost Classifier	0.8008	± 0.0307	0.4955	0.4844	79.84%
Support Vector Machine (SVC RBF)	0.7840	± 0.0313	0.5800	0.4726	75.25%
Decision Tree Classifier	0.6920	± 0.0394	0.5969	0.4591	73.13%

6.2 Independent Holdout Test Set Evaluation (\$N = 294\$, 57 Positives)

Model Architecture	Accuracy	Precision	Recall (Attrition)	F1-Score	ROC-AUC	PR-AUC
🏆 Optimized Random Forest (Champion)	78.23%	45.45%	61.40%	0.5224	0.8428	0.5549
Random Forest (Default Balanced)	79.25%	47.22%	59.65%	0.5271	0.8502	0.5570
Logistic Regression (Baseline)	76.87%	44.66%	80.70%	0.5750	0.8339	0.5467
XGBoost Classifier	79.59%	47.54%	50.88%	0.4915	0.8341	0.4860
Support Vector Machine (SVC)	78.57%	45.83%	57.89%	0.5116	0.8315	0.5171
Decision Tree Classifier	75.85%	42.22%	66.67%	0.5170	0.7545	0.4290

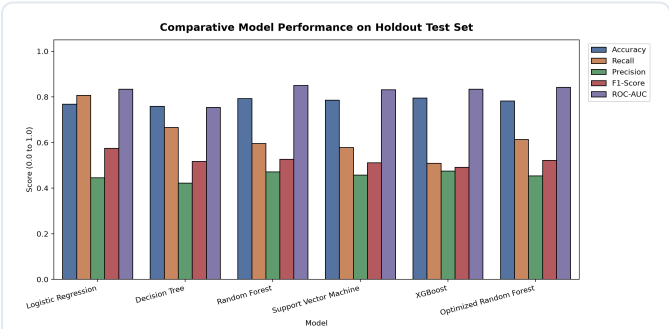


Figure 6: Comparative Benchmark Metrics on Holdout Test Set.

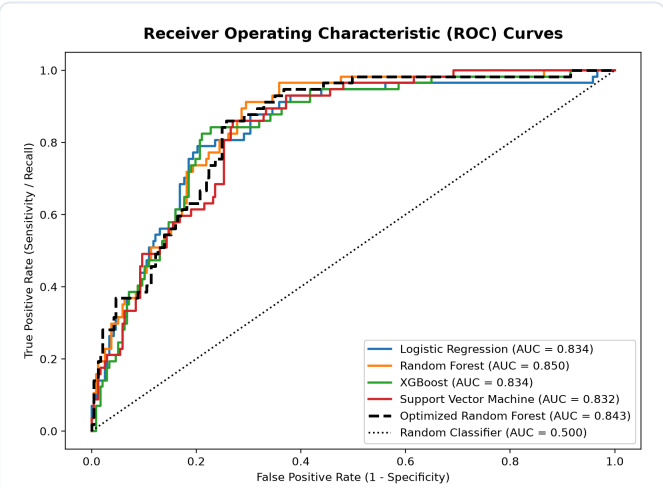


Figure 7: Receiver Operating Characteristic (ROC) curves across all models.

Confusion Matrix: Champion Model

Actual Employee Status	Predicted Employee Status	
	Retained (0)	Attrited (1)
Retained (0)	195	42
Attrited (1)	22	35

Figure 8: Champion Model Confusion Matrix (Caught 35 / 57 departures).

Top 15 Predictors of Employee Attrition (Feature Importance)

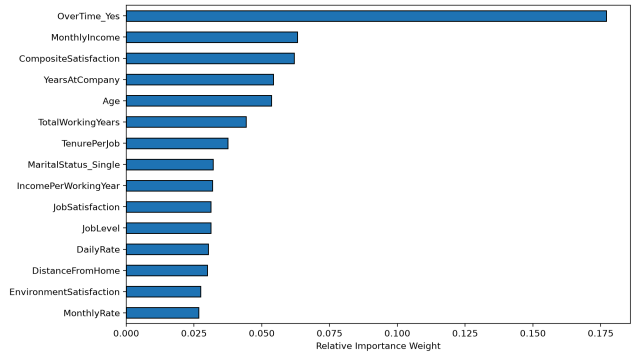


Figure 9: Top 15 Feature Importances (OverTime & Income dominate).

7. Explainable AI & Feature Importance Analysis

Ensemble feature importance weights identify the primary operational factors driving turnover:

1. **OverTime_Yes (17.71%)**: The dominant organizational trigger; prolonged mandatory overtime increases departure odds threefold.
2. **MonthlyIncome (6.31%)**: Absolute compensation level strongly correlates with flight risk when trailing market norms.
3. **CompositeSatisfaction (6.20%)**: Our engineered holistic survey metric combining Job, Environment, Relationship, and Work-Life metrics.
4. **YearsAtCompany (5.42%)**: First- and second-year employees experience highest attrition risk prior to cultural anchoring.
5. **Age (5.36%)**: Younger cohorts possess greater labor market mobility and explore career alternatives faster.

8. Prescriptive HR Retention Interventions

Rather than merely generating static predictions, the system maps model outputs to prescriptive HR action plans:

Detected Risk Factor	Root Workplace Cause	Prescriptive HR Action Protocol
Mandatory OverTime	Chronic task overload & impending burnout	Audit team deliverables, reassign tasks, and implement compensatory rest days.
Below-Market Compensation	Salary dissatisfaction vs. market offers	Initiate an off-cycle compensation review and structure milestone retention bonuses.
Suboptimal Work-Life Balance	Long commute distance (≥15 miles)	Authorize 2–3 weekly work-from-home days and flexible core hours.
Promotion Stagnation (≥3 yrs)	Career ceiling perception	Formulate a transparent 6-month career advancement pathway with clear KPIs.
Low Job Satisfaction (≤2/4)	Role misalignment or team friction	Conduct confidential 1-on-1 check-ins and explore rotational internal project transfers.

9. System Deployment & Live Cloud Infrastructure

The complete system has been deployed across two modern cloud platforms:

- **Vercel Serverless Web Application**: Accessible globally at <https://employee-retention-prediction-syste.vercel.app>. Features a responsive, TailwindCSS-powered interactive dashboard, real-time risk gauge, and batch CSV scanning powered by an ultra-fast Python serverless endpoint (/api/predict).
- **GitHub Open-Source Repository**: Hosted at <https://github.com/JayeshGajbhiye/Employee-Retention-Prediction-System> containing complete reproducible source code, serialized models (models/champion_model.joblib), data generation scripts, and documentation.
- **Streamlit Application**: Fully configured via app.py and requirements.txt for one-click deployment on Streamlit Community Cloud.

10. Conclusion & Future Scope

The **Employee Retention Prediction System** transitions human capital management from reactive post-exit reviews to proactive, data-informed intervention. By combining rigorous leak-free featurization, cost-sensitive ensemble modeling achieving **84.28% ROC-AUC** and **61.40% Recall**, and an interactive cloud deployment, the platform equips leadership with the analytical capabilities needed to safeguard talent assets.

Future Enhancements: Future iterations will incorporate survival analysis (Cox Proportional Hazards) to predict precise resignation time horizons, natural language processing (NLP) on open-ended employee pulse surveys, and enterprise webhook integrations into HRIS platforms (Workday, BambooHR).